

Mark Shperkin

Applied AI Engineer

Contact Info

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About Me

Applied AI Engineer building production-grade AI systems using LLMs, RAG, and agentic workflows to solve real-world business problems. B.S. and M.S. in Computer Science with a focus on AI and deep learning, with hands-on experience designing and deploying scalable ML systems. Proven ability to translate ambiguous requirements into reliable, production-ready solutions, leveraging modern AI tooling for rapid development. Experienced in building end-to-end pipelines, from data ingestion and modeling to deployment. Athletic and military experience built the discipline and teamwork that define my approach to challenges. Fluent in Hebrew, English, and Russian.

Skills

Python, C++, Java, PyTorch, Context Engineering, LLM Evaluation, LLM/RAG Systems, Agentic AI Systems, LangChain, LangGraph, API Integration, NumPy, Pandas, Computer Vision, OOP / OOD, Concurrency / Multithreading, Distributed Systems, Linux, Docker, K8S, GitHub (CI/CD)

Education

- 2024
2025 **M.S. Computer Science**
University of South Carolina
Post-Baccalaureate Certificate in Artificial Intelligence
(GPA 87)
- 2020
2024 **B.S. Computer Science**
University of South Carolina
(GPA 87, Student-Athlete)

Work Experience

Applied AI Engineer RGIS

Israel | 2026

- Designed and deployed AI-powered systems using LLMs and agentic workflows, including agent orchestration and context management, rapidly prototyping and delivering production-ready solutions in a fast-paced, startup environment.
- Led the design and development of a flagship GenAI-driven lead intelligence platform, building an end-to-end agentic pipeline integrating multiple APIs for data retrieval, enrichment, and validation of company and stakeholder data, improving actionable lead quality and volume by 60% while optimizing LLM cost and latency.
- Engineered LLM-driven workflows for data aggregation, reasoning, and scoring, and implemented structured validation and iterative evaluation loops, enabling identification of high-value stakeholders for targeted sales outreach, reducing manual research time to ~15 minutes through simple query-based input.

Software Engineer CleanRate (Startup)

Columbia, SC | 2024 – 2025

- Architected software-first energy-efficiency automation workflows at a startup, including HPXML generation pipelines and automated scoring systems that replaced manual specialist inspection processes with near real-time evaluation.

Teacher Assistant University of South Carolina

Columbia, SC | 2024 – 2025

- Courses Taught: AI; Algorithmic Design II; Programming Languages & Structures
- Developed Java/Python lab modules and automated grading pipelines for 200+ weekly submissions, reducing manual evaluation time by over 90%.

Project Experience

MarkGPT – Personal RAG/LLM Platform

2026

Frontend, Backend, Demo

- Built and deployed a production RAG inference system using FastAPI, ChromaDB vector search, retrieval grounding, scope-restricted prompting, and Anthropic Claude, enabling low-latency semantic retrieval and streaming AI responses.

Master’s Thesis – Multi-Task Deep Learning with Single IMU

2025

ML Pipeline, Labeling Software (Demo), Publication

- Designed and implemented anchor-based temporal detection algorithms, including multi-scale window generation, IoU-based anchor-ground-truth assignment, hard negative mining for class imbalance, and a two-stage 1D non-maximum suppression pipeline to produce precise, non-overlapping activity segments.
- Designed and executed an end-to-end ML pipeline in PyTorch, from dataset collection and labeling to model development and evaluation, achieving ~0.77 micro-F1 for generalizable temporal sensor-based segmentation and classification.

Other Experience

Professional Athlete – Swimming

2004 – 2024

Israeli National Team & University of South Carolina Gamecocks

- Represented the Israeli National Team and USC Division I SEC team, demonstrating elite discipline, teamwork, and high-performance consistency while balancing international competition, military service, and full-time academic commitments.

Military Service – First Sergeant Israel Defense Forces – Air Force

Israel | 2016 – 2019

- Achieved ‘Elite Athlete’ status while balancing international competition with military responsibilities, earning an honorable discharge with a proven record of leadership, reliability, and high-pressure performance.